Report

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2026 / 09 / 04

Summary

General Tasks Status

Task	**Status**	
Correlation Inference Study - Classical Approach	◔	
Correlation Inference Study - Quantum Learning Approach	○	
2026-11 Paper	○	
2026-12 LASNPA Presentation	○	

Correlated Uncertainty Study Tasks Status

Task	Status	Link
Alice ins1222333 Studies		
↪ Uncertainty correlation effect	✓	Plots
RHIC Au-Au/d-Au Studies JETSCAPE		
↪ Uncertainty correlation effect	✓	Plots
Atlas ins1759506 studies		
↪ Uncertainty correlation inference	◑	Notebook
Alice PbPb2760 Scaled Spectra studies		
↪ Uncertainty correlation inference	◔	Notebook
Synthetic data studies		
↪ Uncertainty correlation inference	◔	Notebook

Atlas ins1759506 studies

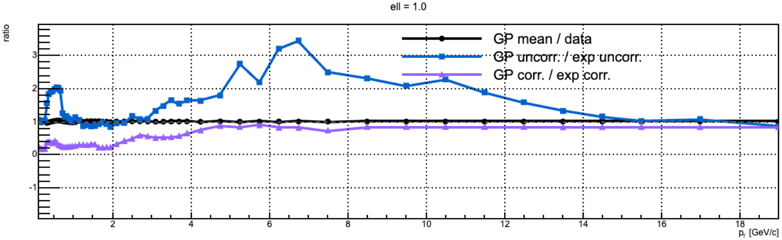

Figure: GP fit: fixed ell = 1.0

Atlas ins1759506 studies

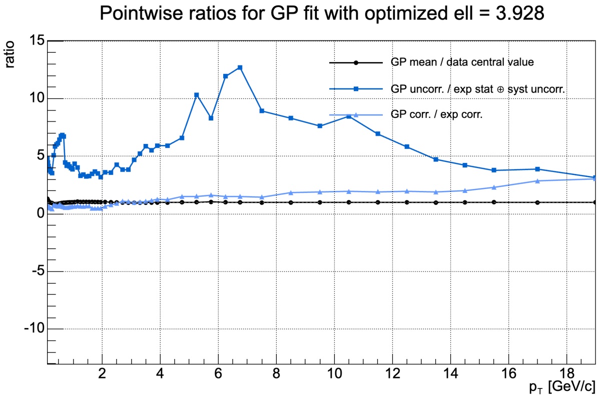

Figure: GP fit: optimized ell

Atlas ins1759506 studies

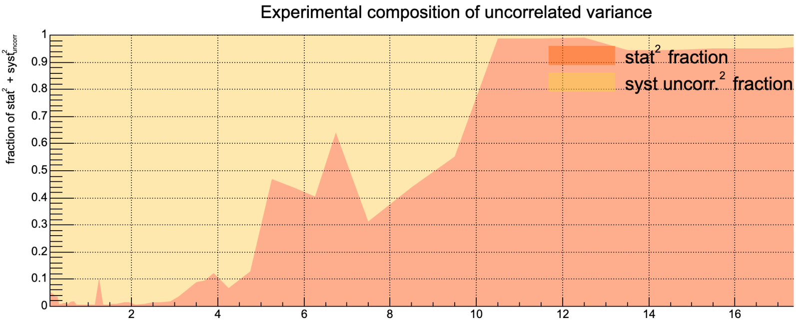

Figure: Ratio between white noise and systematic uncorrelated noise

Alice PbPb2760 Scaled Spectra studies

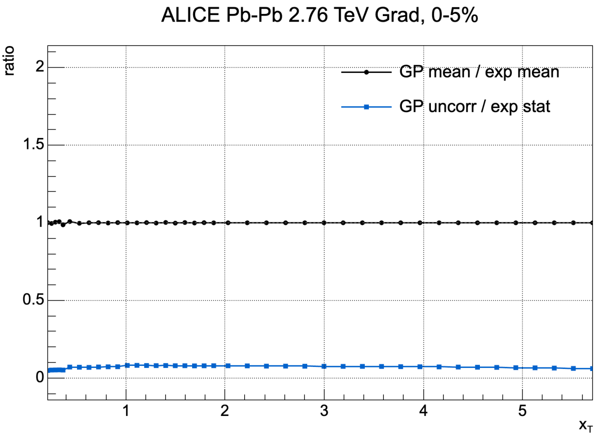

Figure: GP fit: fixed ell = 0.2

Alice PbPb2760 Scaled Spectra studies

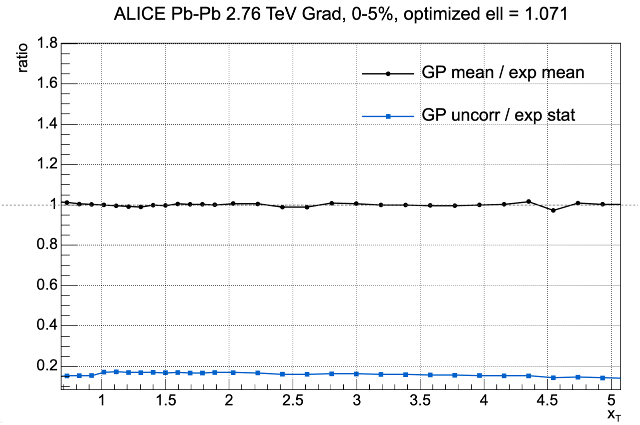

Figure: GP fit: optimized ell

Selected Article

Responses of multiparticle observables to multidimensional nuclear deformation in relativistic heavy-ion collisions

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Relativistic heavy-ion collisions provide a unique opportunity to probe ground-state nuclear structure through its imprint on the initial collision geometry. We investigate how multiparticle observables respond to combined variations of quadrupole deformation, triaxiality, and hexadecapole deformation, using ^{129}Xe+^{129}Xe collisions as a representative testing ground. We perform a joint analysis in the three-dimensional $(\beta_2, \gamma, \beta_4)$ parameter space and construct initial-state estimators for several flow and mean transverse momentum correlation observables. At the initial-state level, ρ_2 is primarily sensitive to β_2 and γ, with its sensitivity to γ enhanced at nonzero β_2. The nonlinear response coefficient $\chi_{4,22}$ is predominantly sensitive to β_4, while its dependence on β_2 and γ remains comparatively weak. Higher-order correlators exhibit more complex multidimensional response patterns; in particular, ρ_{224} shows a dependence on γ and β_4 that becomes more pronounced at finite β_2. We further employ the iEBE-VISHNU hybrid model to examine whether these deformation sensitivities survive the subsequent dynamical evolution. The final-state calculations indicate that the sensitivity of ρ_2 to β_2 and γ is largely preserved, whereas the β_4 sensitivity of $\chi_{4,22}$ is substantially reduced. For the other higher-order observables, the initial-state sensitivities are modified by the evolution or cannot be resolved with the present statistics.

Figure: Multidimensional nuclear structure deformation study.